

Jaime Andrés Castañeda

Data scientist

Colombia

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Portfolio: <https://jacasta2.github.io>

PROFILE

Computer Science Engineer, Master in Systems Engineering and PhD in Economics. Has extensive experience leading and working on experimentation and data science projects to strategically support decision-making and drive business results. Has developed experimentation and data science apps, supervised and unsupervised machine learning models using Python and SQL, Power BI dashboards and statistical analyses using Python, Stata and SPSS to support decision-making and the development of improved business strategies. Skilled in Python, SQL, Machine Learning, NLP, Pandas, Numpy, Matplotlib, Seaborn, Scikit-learn, Keras, Streamlit, FastAPI, VS Code, Git, Databricks, Postgres, Power BI, Stata and SPSS. Interested in data science and machine learning roles with career development as a technical leader.

PROFESSIONAL EXPERIENCE

S&P Global

Data Scientist

05.2025 – Present

AI and Machine Learning research in Energy, Agriculture and Climate for S&P Global Commodity Insights

Procalidad Analytics

Senior Data Scientist

04.2024 – 05.2025

- Perform extract, transform and load (ETL) processes that support business products
- Construct features that feed downstream machine learning models and business prediction initiatives
- Build machine learning models for long-term predictions that support strategic decision-making

Achievements

- Led the construction of frost metrics using Python and Spark, complementing the weather features construction pipeline that feeds agricultural yield prediction models
- Delivered ETL processes for multi-country crop data integration using Python and Spark, improving datasets to enhance the performance of predictive agriculture models
- Built and fine-tuned deep learning models to forecast long-term yield outlooks for key global crops in multiple climate change scenarios, powering a client-facing analytics product

Ministry of Information and Communication Technologies of Colombia

Data Scientist

03.2023 – 12.2023

- Implement extract, transform and load (ETL) processes and components that support data science initiatives
- Develop data science and machine learning products to support decision-making in different areas and drive business results
- Manage data science projects addressing data collection, preparation and storage and its exploitation for evaluating strategies and value generation

Achievements

- Led the construction of a composite index in Python with more than 40,000 school records to support the Internet connectivity prioritization policy
- Developed Python code to clean and extract data from legal documents using regular expressions and ChatGPT through the OpenAI API
- Built 5 dashboards using Power BI to track the effectiveness of Internet connectivity and Information Technology training projects

- Created 2 updated map templates of Colombian towns and departments using Python and Power Query for the standardization of Power BI dashboards

Correlation One

Data Science Fellow

03.2022 – 07.2022

- Identify relevant data sources and collect the data necessary to address business problems using tools such as Python and its interaction with APIs
- Perform data preparation processes (cleaning, transformation and feature engineering) to ensure data quality and integrity
- Build and train predictive models using data science and machine learning techniques to support decision-making

Achievements

- Performed an extract, transform and load (ETL) process with more than 1.5 million traffic accident records using Python, SQL and Postgres
- Built heatmap visualizations using Python, SQL and Seaborn to analyze motorcycle accident risks, identifying that more than 50% of accidents occur with passenger vehicles and other motorcycles
- Developed an unsupervised machine learning model using Python and SQL that clusters highway corridors for road safety operations, prioritizing corridors that total close to 50% of accidents and more than 40% of fatalities

ADDITIONAL EXPERIENCE

Universidad del Rosario, *Associate Professor* 2021 – 2022

Universidad del Rosario, *Principal Professor* 2016 – 2021

Massachusetts Institute of Technology (MIT), *Postdoctoral Fellow* 2014 – 2015

EDUCATION

Università della Svizzera italiana, *PhD in Economics* 2009- 2013

Universidad Nacional de Colombia, *Master's in Systems Engineering* 2008 - 2009

Universidad Nacional de Colombia, *Bachelor's in Systems and Informatics Engineering* 2003 - 2007

SKILLS

Technical skills: Data science, Machine Learning, Analytics, Natural Language Processing (NLP), Extract Load Transform (ETL), Web scraping, Data wrangling, Data visualization, Pandas, Numpy, Matplotlib, Seaborn, Scikit-learn, Keras, Streamlit, FastAPI, VS Code, Git, Databricks, Postgres, Power BI, Stata, SPSS, Excel, Qualtrics
Soft skills: Rigor, Organization, Teamwork, Proactivity, Planning, Communication